THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Rolling friction is less than sliding friction. Reason (R): The contact area is smaller and no sliding occurs in rolling. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A man in a boat pushes against the shore with a pole. The boat: 1

(a) moves to the shore **(b)** rotates only
(c) stays **(d)** moves away from the shore

3. A 1000 kg car pulls a 500 kg trailer with an acceleration of 1 m/s squared. The tension in the coupling (no friction) is: 1

(a) 500 N **(b)** 1500 N **(c)** 250 N **(d)** 1000 N

4. A body moving with a velocity of 4 m/s is acted on by a force opposite to its motion producing a retardation of 2 m/s squared. How far does it go before stopping? 1

(a) 4 m **(b)** 8 m **(c)** 16 m **(d)** 2 m

5. Assertion (A): A gun recoils on firing a bullet. Reason (R): The bullet exerts an equal and opposite force on the gun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A body of mass 2 kg moving at 3 m/s is hit by a force of 4 N for 2 s in the direction of motion. Its new velocity is: 1

(a) 1 m/s **(b)** 4 m/s **(c)** 11 m/s **(d)** 7 m/s

7. A force F gives an acceleration a1 to mass m1 and a2 to mass m2. If both are joined, the acceleration is: 1

(a) (a1 + a2)/2 **(b)** a1 - a2 **(c)** a1 + a2 **(d)** a1 a2/(a1 + a2)

8. A cart of mass 20 kg is pulled with a net force of 50 N for 4 s. Its final velocity from rest is: 1

(a) 20 m/s **(b)** 10 m/s **(c)** 5 m/s **(d)** 2.5 m/s

9. **Assertion (A):** A gun recoils backward when a bullet is fired from it. 1
Reason (R): The total momentum of the gun and bullet is zero before firing and remains zero afterwards, so the gun gets momentum opposite to that of the bullet.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Rolling friction is less than sliding friction. 1
Reason (R): Friction always opposes the relative motion between surfaces in contact.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a wheel come to rest when a motor is switched off? Which force is responsible? 2

12. A man of 60 kg jumps down from a height of 5 m and stops in 0.2 s. Find his speed on landing and the average force (g = 10). 2

13. A 2 kg trolley moving at 6 m/s is stopped in 3 s by a constant force. Find the force and the stopping distance. 2

14. A man pushes a wall with 100 N. What force does the wall exert on him? Does the wall move? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the air resistance on a falling parachutist changes with speed and how terminal velocity is reached. Find the terminal air resistance for a 70 kg man. 3

16. A 0.15 kg cricket ball bowled at 30 m/s is hit back along the same line at 40 m/s with contact time 0.01 s. Find the impulse and the force. 3

17. Explain how the effect of a given force on the motion of two bodies depends on their masses with F = 12 N on masses 2 kg and 6 kg. 3

OR

A mass of 5 kg hangs from a rope. Find the tension when it 3
(a) is at rest
(b) moves up at 2 m/s squared
(c) moves down at 2 m/s squared (g = 10).

18. A stone tied to a string is whirled in a circle and the string breaks. Explain the path of the stone afterwards using the law of inertia. 3

OR

The momentum of a body of mass 3 kg changes from 12 kg m/s to 24 kg m/s in 4 s. Find the force, the initial and final velocities. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A rifle of mass 4 kg fires a bullet of mass 20 g at a speed of 200 m/s. The shooter feels the rifle push back against the shoulder. 5
(i) What is the total momentum of the rifle and bullet before firing?
(ii) Find the momentum of the bullet.
(iii) Find the recoil velocity of the rifle.
(iv) Explain why the speed of the rifle is much smaller than that of the bullet, and state its direction.

20. A block of mass 10 kg on a rough floor is pulled by 60 N and the friction is 20 N. Find the acceleration, the velocity and the distance after 5 s, and the momentum then. 5

OR

A car moving at 20 m/s is stopped by a constant braking force of 5000 N. If the mass is 1000 kg, find the retardation, the stopping distance and the time. How do these change if the speed is doubled? 5

END OF QUESTION PAPER — **Total: 40 Marks**